

The add-ons are vulnerable

Any kind of third-party applications, such as flash, quick time, java are targets for finding exploits

Social engineering

Targeted attacks focus more on fooling the user than exploiting a weakness in the computer itself

Conficker havoc

Although the conficker worm is more or less neutralized, it spread most in the Asian subcontinent, speaking volumes of our security

Feature

Securely deleting data

Ideally, you should be using public computers like a ghost. No trace of your activities on the computer should be left behind. If you have downloaded any kind of data, or have created data in the cybercafé, it is a good idea to delete the data before you leave. Even if you use [Shift] + [Delete] to get rid of the data, or empty the Recycle Bin, the data is still there on the hard drives, but not indexed in the filesystem. This means that fairly simple and commonly available file recovery software can be used to recover the deleted data. There are a number of free and portable programs to securely delete data from the hard drive. These programs manage to do this by overwriting the empty space, or the areas of the hard disk occupied by the file. We will be using two programs for securely deleting data – Free Commander and Eraser. Free Commander is faster, less secure, but should do for most day to day purposes. Eraser is a highly-specialized application, and can offer as much security as time permits.

Free Commander is a free file browser that makes exploring the file system easier. Folders show up with sizes, and there is a dual-pane interface that makes moving data around a breeze. To securely delete data, select the files or folders, then go to File > Wipe.

Free Commander offers upto ten overwrite runs. Each “run” involves rewriting the area where the data was stored with random data. The more the runs, the more irrecoverable the data becomes. However, if you have deleted data, without wiping it, then it can be recoverable by a file recovery software. Free Commander has no function to erase the empty hard disk space, and rewrite the empty information. This is where Eraser steps in.

Eraser is a deceptively simple looking program. Go to File and add tasks to add the parts of the hard disk that has to be erased. The user can add empty hard disk space, specific folders, or individual files. Each operation is called a task, and any number of tasks can be added to a list known as the task list. Eraser goes through the task list, erasing the identified data one at a time. At each instance of an erasing operation, there are a whole range of secure deletion

options available to the user. There are a few default patterns in which the erasing occurs, but this can be entirely tweaked by the user. The most secure default pattern offered by the program is the Gammunn method, which involves thirty five passes. This is more than anyone really requires. Click on new to create your own pattern. You can specify an unlimited number of passes, and define what kind of data is used to overwrite the file or empty space in

each of these passes. A pattern overwrite uses a specified pattern, and the pseudorandom data overwrites with random characters. We specified an operation which implements 101 passes.

Note here, that the more passes you specify, the more is the time taken. Overwriting files and folders is a relatively fast operation, compared to rewriting all the empty space in a hard disk. Even a 10 passes operation will take a long time when it comes to clearing hard disks

with a lot of empty space. Also note that two or more identical passes, following the same pattern (say all zeroes) is the same as one pass with that pattern. Either alternate the pattern, or sandwich patterns between two pseudorandom data passes.

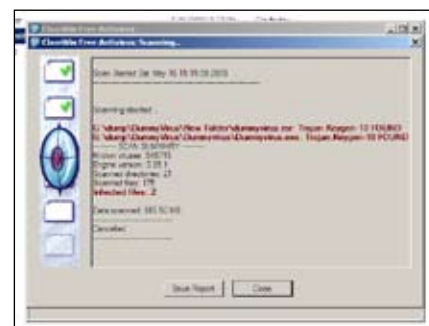
Protect your USB drive from viruses

The first thing to do when you plug in your USB drive into a public computer is to identify and disable malicious processes running on the host computer. A great utility for doing thing is Process Explorer.



Stop Trojan Infections

A simple way to disable any virus, worm or trojan from entering your USB drive, is to create a folder by the name of “autorun.inf”. Once this is done, right click on the folder, navigate to properties, hide the folder as well as write protect it. This works as well with a text file of the same name. This ensures that even if a virus, trojan or worm does get into the USB drive, it does not being to operate as soon as the drive is connected to any computer.



Clamwin identifying Trojans in the USB drive

Process Explorer is a free alternative to Windows Task Manager. Process Explorer can halt or disable undesirable processes. The most useful little feature in Process Explorer is, if there is a process that restarts when you stop it, you can suspend in and leave it hanging while you continue with your work.

Some experience with the process manager is required before you can identify the harmful or irregular processes right away. The website ProcessLibrary.com lists all the harmful processes, and has a listing of the most common threats. If there is a process you are suspicious about, this is the place to identify it. ProcessLibrary also has lists of legitimate processes, so you know at any point of time everything that is going on in your computer.

If you want to use more active measures, Clamwin is a free anti virus software that is also portable. As with all anti virus software, the anti virus definitions have to be updated regularly. This can be done when you run the anti-virus from the cybercafé itself. The first time you update the definitions, the process is going to be quite lengthy, as the installer of the anti-virus does not come with definitions. The updates are much faster after the first run.

Scanning for viruses, worms and Trojans takes some time. Not only can you prevent viruses from infecting your USB drive, you can also scan and fix the host computer from an installation in your drive.

These software should help you stay secure in most situations, and all of them are in the accompanying DVD. However, attackers are likely to target the humans as much as the computers. Don't give away banking details to anonymous people, no matter who left you property in their wills, or which lottery you won. Don't follow links to banking sites through e-mails, as they may be dummy sites that are made to look like the official sites. Use the URL of the official sites that you trust, and have used before.